034IN -FONDAMENTI DI AUTOMATICA - FUNDAMENTALS OF AUTOMATIC CONTROL

FULL WRITTEN EXAMINATION - A.Y. 2023/2024

January 20, 2025

PART 1 - Problem 1- Solving an ODE using the Laplace Transform

Tag List: `[STANDARD ASSESSMENT] [SYMBOLIC]`

Using the **Laplace transform** find the **causal function** $y(t)$ solution for the following Ordinary Differential Equation (ODE):

$$\frac{\mathrm{d}^2 y(t)}{\mathrm{d}t^2} + 4\, y(t) = \cos(t) \cdot \mathbf{1}(t)$$

As usual, $\mathbf{1}(t)$ denotes the **unit step function** (the Heaviside function): $\mathbf{1}(\tau) = \begin{cases} 0 & \longleftrightarrow & \tau < 0 \\ 1 & \longleftrightarrow & \tau \geq 1 \end{cases}$

Assume as **initial conditions** (ICs) the following:

$$\begin{cases} y(0) & = & +\frac{4}{3} \\ \left.\frac{\mathrm{d}y(t)}{\mathrm{d}t}\right|_{t=0} & = & +2 \end{cases}$$

Assignment

Your solution has to:

- Use the Matlab's **Symbolic Toolbox**.
- Solve the ODE in the Laplace transform domain, taking into account the assigned initial conditions.
- Evaluate the inverse Laplace transform of the solution, making use of the available Symbolic Toolbox commands and functions .
- Assign the solution in the time domain to the **symbolic** variable **y_sol**.

Solution

```
syms t s
syms y(t) u(t)  % the variables in the time domain
syms Y  % the Laplace transform

assume(t >=0)

% ----------- ICs -----------------------------------------
```

```
y0VAL =    sym(4/3);   % the initial conditions: the value y(0)
dotY0VAL = sym(+2);   % the initial conditions: the value dot_y(0)

% ------ the time derivatives of the function y(t) ------
dot_y = diff(y,t); % the 1st derivative of y
ddot_y = diff(dot_y,t);
% -------------------------------------------------------

% ---- the ODE -----
eqnODE = ddot_y +4*y == cos(t); % the ODE to be solved
% ---- the ODE -----

eqnLT = laplace(eqnODE, t, s); % applying Laplace transform

eqnLT1 = subs(eqnLT,laplace(y,t,s),Y);
eqnLT2 = subs(eqnLT1, y(0), y0VAL);
eqnLT3 = subs(eqnLT2, subs(diff(y(t), t), t, 0), dotY0VAL)
```

eqnLT3 =

$$Y\,s^2 - \frac{4\,s}{3} + 4\,Y - 2 = \frac{s}{s^2+1}$$

```
Ysol = simplify(solve(eqnLT3, Y))
```

Ysol =

$$\frac{\frac{4\,s}{3} + \frac{s}{s^2+1} + 2}{s^2+4}$$

```
y_sol = ilaplace(Ysol,s, t)
```

y_sol =

$$\cos(2\,t) + \sin(2\,t) + \frac{\cos(t)}{3}$$